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A Tertiary Study on Microservices: Research Trends and Recommendations

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Abstract—The development and adoption of microservices, as one of the most promising directions for developing heterogeneous distributed software systems, have been driven by dynamic changes in business and technology. In addition to the development of new applications, a significant aspect of microservices is the migration from legacy monolithic systems to microservice architectures. Such development trends are accompanied by an increase in the number of primary and secondary publications addressing microservices, highlighting the need to systematize research at a higher level. The objective of this study is to comprehensively analyze secondary studies in the field of microservices from the following five aspects: (1) publishing trends, (2) quality trends of secondary studies, (3) research trends, (4) domains of implementation, and (5) future research directions. The study follows the guidelines for conducting a systematic literature review. The findings were derived from 44 secondary studies published in the period from January 2016 to January 2023. These studies were organized and analyzed to address the five proposed research questions pertaining to the study objectives. The findings suggest that the most promising research directions are related to the development, implementation, and validation of new approaches, methods, and tools that encompass all the phases of the life cycle. Additionally, these research directions have applications in a variety of business and human life domains. Recommendations for further literature reviews relate to improvement of quality assessment of selected studies, more detailed review of architecture quality attributes, inquiry of human factor issues, and certain maintenance and operation issues. From the methodological aspect, recommendations relate to using social science qualitative methods for more detailed analysis of selected studies, and inclusion of gray literature that will bring the real experience of experts from industry.

Keywords: microservices, tertiary study, systematic literature review

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1. INTRODUCTION

Microservices have become immensely popular and widely used architectural model for cloud-based applications. They can be considered successors to service-oriented architectures (SOA) and represent the new trend in developing distributed software systems [1, 2]. These small, independent services offer improved performance and support for continuous delivery [3]. While there is no clearly defined principle for breaking up monolithic applications, certain factors should be considered such as the number of objects owned by a service, the level of responsibility, and the team distribution [4]. Initially, microservices were adopted by large companies like Amazon, LinkedIn, and Netflix, and later by other companies [5, 6]. This led to an increasing trend in the use of microservices for developing cloud-based applications. According to Larrucea et al., the adoption of microservices in developing or reengineering software systems requires embracing a new organizational and business culture, with special attention to system details and architecture [7]. Additionally, the adoption of DevOps is crucial for better integration throughout the system life cycle [8]. Furthermore, the development of new complex and heterogeneous software systems based on microservices necessitates the adoption of agile approaches not only in development but also in deployment, operation, and monitoring [6]. In this

context, the utilization of contemporary life cycle approaches based on DevOps and continuous software engineering (Continuous Integration, Continuous Delivery, Continuous Deployment, and Continuous Monitoring) becomes necessary for the success and competitiveness of software organizations [9].

Literature reviews, such as systematic literature reviews (SLRs) [10] or systematic mapping studies (SMSs) [11, 12], have recently been embraced by the software engineering research community to systematically analyze evidence on practice. This recognition has led to the emergence of Evidence-Based Software Engineering (EBSE) as a concept applicable to both researchers and practitioners [13]. In addition to reviewing academic literature, commonly referred to as "white literature," these reviews also encompass "gray literature" sources such as blog posts, white papers, industrial magazines, and videos. Consequently, multivocal literature reviews, which rely on both white and gray literature, have gained prominence [14].

Given the aforementioned context, the objective of this study is twofold: (a) to present the current state-of-the-art in microservice research and (b) to propose recommendations for further research. Special emphasis is placed on the quality of the published studies and the prevailing trends and research landscape in microservice implementation. To achieve these objectives, a systematic literature review was conducted following the guidelines proposed in [10, 13]. The search and selection process yielded 44 secondary studies, forming the foundation for drawing research findings. Based on these findings, we discuss implications, validity, and recommendations for further research.

This paper is structured as follows. Section 2 provides background information on microservices and tertiary studies related to microservices. Section 3 outlines the research methods employed in the study, while Section 4 presents the research findings. In Section 5, we delve into the research implications, validity, and recommendations for future reviews. The concluding section summarizes the key insights and highlights avenues for further research.

2. BACKGROUND

In this section, we provide an overview of the fundamental concepts of microservices in the first subsection. The second subsection offers a concise review of published tertiary studies about microservices.

2.1. Microservices

Microservices have recently emerged as an immensely popular software architecture paradigm. They consist of multiple components that collectively form the entire system. Each component performs a single task, with its boundaries shielding it from external knowledge, while the processed results can be shared and accessed by other microservices [2]. This structure ensures stability, even when system upgrades or expansions are necessary. With microservices, clients can be confident that any changes or growth in their business will be seamlessly integrated into the software, as these applications are designed to support and facilitate work processes. Microservices have proven to be superior to monolithic architectures, particularly in terms of meeting business requirements, ensuring systems reliability, enhancing maintainability, and bolstering infrastructure resilience. Even though they require more teams and greater effort, the long-term benefits make the investment worthwhile [15]. Presently, it has become a common practice to migrate monolithic or legacy systems to service-oriented architectures [16], particularly microservice architectures, which significantly improve overall system performance [17, 18]. As the field of microservices engineering matures, the utilization of specific design patterns will become commonplace, resulting in more efficient development practices and better fulfillment of various architectural quality attributes [19].

Microservices-based systems consist of numerous individual microservices, each performing a specific functionality, with no dependencies among them. Consequently, if one component fails, the entire system remains unaffected. The principle of *Autonomy* is responsible for this behavior, alongside other key principles, which include [15]: *Resilience*—ensuring that the application can continue providing services even if a specific microservice encounters failure; *Transparency*—exposing the necessary details and providing documentation for each microservice, enabling other services to effectively utilize it and understand its current state; *Automation*—employing tools that enhance the efficiency, reliability, and scalability of the microservices' building and maintenance processes; and *Alignment*—emphasizing the significance of harmonizing different microservices within the system, as well as aligning them with the organization's goals and needs. By adhering to these key principles, microservices architectures can achieve robustness, flexibility, and effective collaboration within the system.

Microservices offer another advantage through utilization of APIs for communication. This approach enhances the flexibility and operability of microservices by employing a universal languages that are independent of the programming languages used for coding the application. Given the multitude of components in microservice-based systems, the selection of a communication method becomes crucial. Newman discusses various communication styles, such as synchronous blocking, asynchronous non-blocking, request-response, event-driven, and common data, along with their respective advantages and disadvantages [2]. The choice of communication style depends on the specific requirements of the tasks at hand. However, the most commonly employed communica-

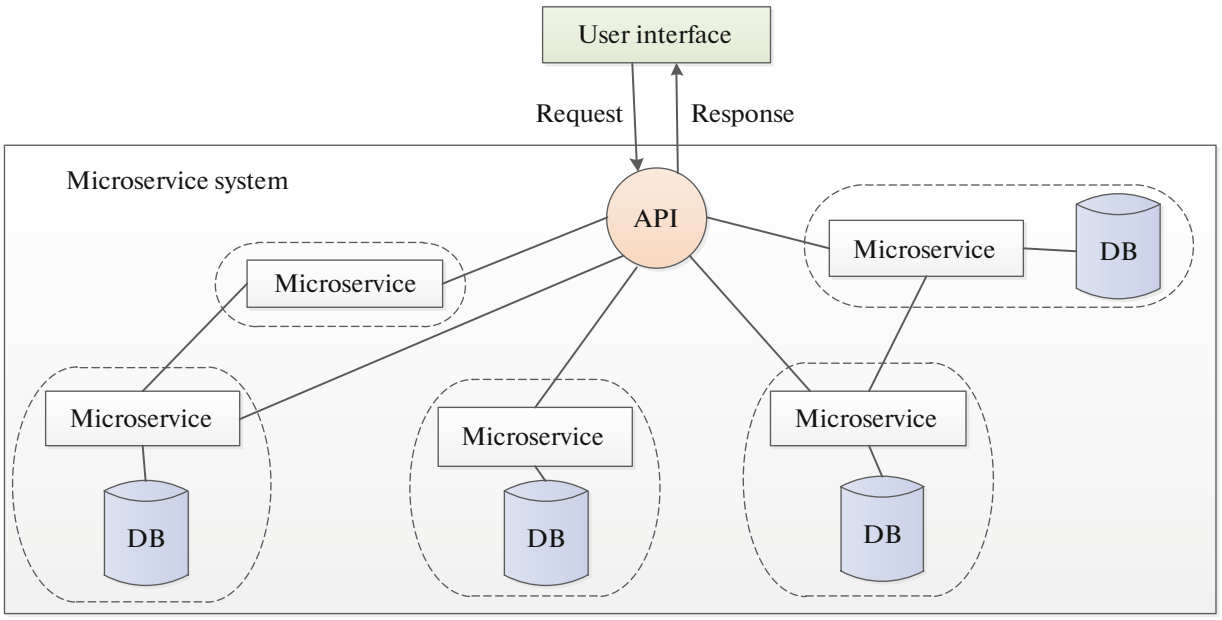


Fig. 1. Microservices-based system.

tion mechanism is the HTTP protocol, which aligns with the REST architectural style [20]. This choice ensures compatibility and ease of integration within microservices architectures.

Figure 1 illustrates a typical example of microservice architecture. An integrated microservice system comprises a user interface and a collection of microservices (although some microservices may not have a user interface). The number of microservices can vary depending on the specific purpose of the system. Unlike monolithic systems, microservices operate with individual databases; each service has its dedicated database. To interact with the system, users send requests through the user interface in a format defined by the API. The API receives each request and forwards it to the corresponding microservice responsible for resolving it. Once the API receives a response from a selected microservice, it relays it back to the user. Similarly, APIs serve as the communication channels for microservices to interact with one another and share data. This enables seamless coordination and collaboration among the microservices within the architecture.

Microservice organizations encompass various types, with one of the most prevalent being container-based architecture [21]. Containers play a vital role in physically isolating individual microservices from one another, as microservices operate within distributed systems. By leveraging containers, the requirement for deploying microservices on dedicated machines to achieve separation is eliminated. Containers provide the necessary isolation between microservices, reducing the reliance on physical machines for deployment [2].

2.2. Related Work

Tertiary studies in software engineering serve the purpose of reviewing secondary studies and conducting meta-analyses on specific research topics. Recent tertiary studies have covered a range of subjects including DevOps [22], architecting systems of systems [23], cloud computing [24], agile software development [25], variability in software product lines [26], and testing artifact quality [27].

During the search conducted in December 2022, two tertiary studies on microservices were identified. The first study, titled "Research on Microservice Architecture: A Tertiary Study" by Liu et al. [28], and the second study titled "Microservice Architecture: A Tertiary Study" by Costa et al. [29], both primarily focus on microservices architecture. However, other aspects such as migration to microservices, security issues, or implementation in specific sociotechnical contexts are not directly addressed. Table 1 provides comprehensive information on these tertiary studies, including the period covered and the number of secondary studies included in each review. The review types encompass systematic literature reviews (SLRs), systematic mapping studies (SMSs), systematic grey literature reviews (SGLRs), and multivocal literature reviews (MVRs).

Liu et al. (2022) conducted a systematic literature review, identifying 37 secondary studies on microservices published between 2016 and 2021 [28]. Liu et al. formulated two research questions for their review: (RQ1) What are the common topics addressed in secondary studies related to microservices architecture (MSA), and what are their findings? (RQ2) What are the potential areas for new research in the field of

Table 1. Tertiary studies on microservices

Study reference	Time span	SLRs	SMSs	MVRs	SGLRs
Liu et al. (2022) [28]	2016–2021	17	20	0	0
Costa et al. (2020) [29]	2016–2019	5	14	2	1

MSA? To assess the quality of the secondary studies, the authors have employed the DARE quality criteria as described in [13].

Costa et al. (2020) conducted a systematic literature review, identifying 22 secondary studies on microservices published from 2016 to 2019 [29]. The original study was written in Portuguese, necessitating the translation of methodological issues and results into English during the analysis. Costa et al. addressed the following research questions: (RQ1) Which secondary studies have been published in the field of microservices? (RQ2) What research topics on microservices have been investigated? (RQ3) What emerging patterns have been identified? (RQ4) What solutions and support tools have been utilized to facilitate the development and operation of microservices architecture? (RQ5) In which areas, particularly in the industry, are microservices being applied? (RQ6) Which topics exhibit gaps and require further exploration in future microservices research? Costa et al. assessed the quality of secondary studies using the DARE quality criteria as described in [13].

3. RESEARCH METHOD

This study adheres to the guidelines outlined for conducting systematic literature reviews, as proposed in [10, 13]. The research process is structured into three primary phases: (1) planning the review, (2) conducting the review, and (3) reporting the findings.

3.1. Planning the Review

The planning phase includes justifying the need for a tertiary study, determining research questions, selecting digital databases, and defining the studies search and selection process with clearly stated keywords for searching, inclusion/exclusion criteria, and quality assessment criteria. These issues are discussed in the following subsections.

3.1.1. Need to conduct a tertiary study on microservices. Microservices have emerged as a significantly popular research field, witnessing a substantial growth in the number of studies presenting theoretical and empirical findings over the past decade. Due to this development, the consolidation of individual study outcomes has become essential, leading to the execution of systematic reviews. These reviews encompass various approaches, such as systematic literature reviews [10], systematic mapping studies [11], and even multivocal literature reviews that encompass gray literature [14].

During the search conducted in digital libraries, two articles were identified as tertiary studies on microservices: a study titled "Research on Microservice Architecture: A Tertiary Study" by Liu et al. [28], and a study titled "Microservice Architecture: A Tertiary Study" by Costa et al. [29]. It is worth noting that [29] is written in Portuguese, while [28] is in English.

Insights derived from these two studies revealed a consistent increase in the number of secondary studies over the years. This trend is expected to continue in 2023 and beyond, highlighting the need for a new reviews of recent secondary research. Furthermore, these two studies did not specifically investigate quality trends in secondary studies, migration-related issues, or implementation challenges in various domains. Therefore, further research opportunities exist in these areas. Considering these factors, conducting a new systematic literature review that incorporates the most recent studies and focuses on assessing the quality of secondary research would greatly contribute to enhancing the knowledge base within the field of software engineering.

3.1.2. Research questions. To enhance comprehension of the current research landscape in the field of microservices, several research questions have been formulated to explore the findings of secondary studies, trends in the quality of secondary studies, and challenges that warrant future research. The proposed research questions (RQs) are as follows.

RQ1. What are the publishing trends observed in secondary studies?

RQ2. What is the trajectory of quality trends in secondary studies?

RQ3. What are the predominant topics investigated in secondary studies?

RQ4. In which domains are microservices commonly implemented?

RQ5. What future research directions have been identified?

By addressing these research questions, a comprehensive understanding can be obtained regarding the publication patterns, quality aspects, research topics, implementation domains, and future research directions within the realm of microservices.

3.1.3. Selection process of secondary studies. We have carefully selected specific keywords to encompass the concepts of microservices and secondary studies that can be conducted as systematic literature reviews

or systematic mapping studies. To this end, we have divided the keywords into two groups: the first group includes the keywords "microservices" and "microservices architecture," while the second group comprises the keywords "SLR," "Systematic literature review," "SMS," and "Systematic mapping study." To perform searches in digital databases, we utilized the following search strings:

[1]: ("microservices architecture" OR "microservices") AND ("SLR" OR "Systematic literature review")

[2]: ("microservices architecture" OR "microservices") AND ("SMS" OR "Systematic mapping study")

The process of searching and selecting studies involved the following phases (Ph#no.):

• Ph#1: Searching digital libraries using search strings composed of relevant keywords;

• Ph#2: Selecting specific studies based on their title, abstract, and keywords. This phase also involves removing duplicates (in case a study appears in multiple databases), and selecting the most recent version of the study (if there are multiple versions by the same authors);

• Ph#3: Employing the snowball method [30] to search for additional studies and minimize the possibility of overlooking or omitting any relevant secondary studies;

• Ph#4: Applying inclusion/exclusion criteria to the studies selected in the phases Ph#2 and Ph#3;

• Ph#5: Conducting a detailed reading and analysis of the studies that passed Ph#4, while comparing them against defined quality assessment criteria.

The digital libraries used for searching secondary studies include ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, Wiley Online Library, and MDPI. These libraries were selected because they publish a majority of the leading journals and conference proceedings in the field.

The following inclusion and exclusion criteria were applied to filter the studies obtained from the search of these digital libraries.

• Inclusion criteria.

◦ **I1**. A study reviews relevant studies on microservices.

◦ **I2**. A study follows guidelines for conducting SLR or SMS.

◦ **I3**. A study answers research questions in the domain of microservices.

• Exclusion criteria.

◦ **E1**. Full text of a study is not available.

◦ **E2**. A study is not peer-reviewed.

◦ **E3**. The study is less than 6 pages.

◦ **E4**. A study is not written in English.

◦ **E5**. A review study that includes gray literature.

All secondary studies identified through the systematic search using search strings or the snowball method are evaluated against these inclusion and exclusion criteria. If a secondary study fails to meet even one criterion, it is excluded from further analysis.

3.1.4. Quality assessment of secondary studies. The secondary studies were evaluated for quality based on guidelines proposed in [13]. The primary objectives of the quality assessment were as follows: (1) to identify and exclude low-quality studies from the detailed analysis and synthesis of review findings and (2) to provide insights for addressing RQ2 regarding quality trends in secondary studies. A three-point scale was used for the quality assessment, with values of 1 (Yes), 0.5 (Partly), and 0 (No). This scale was based on the five questions (Q2–Q6) proposed in [13], along with an additional question concerning the use of the review methodology (Q1). The questions were as follows.

Q1. Is the review methodology clearly stated and appropriate?

Q2. Are the review's inclusion and exclusion criteria described and appropriate?

Q3. Is the literature search likely to have covered all relevant studies?

Q4. Did the reviewers assess the quality/validity of the included studies?

Q5. Were basic studies adequately described?

Q6. Were the extracted data from included studies synthesized in the findings?

The authors of this paper have agreed that studies with an average quality score of less than 0.5 will be excluded of the detailed analysis.

3.2. Conducting Review

The search for secondary studies commenced in January 2023. The first phase (Ph#1), which involved searching selected digital libraries using the provided search strings, has concluded, resulting in the identification of 821 papers that will proceed to subsequent review phases. The details of the search conducted in digital libraries are presented in Table 2.

Figure 2 illustrates the phased process of selecting secondary studies, while Table 3 provides specific details regarding the selection of SLRs and SMSs throughout the phases.

The fourth filtering phase (Ph#4) involves assessing studies against the proposed inclusion/exclusion criteria. After evaluating each of the 57 studies, 44 were selected for further analysis and quality assessment. Three studies were excluded based on the E3 exclusion criterion, one study based on the E1 exclusion criterion, and nine studies based on the E5 exclusion criterion.

3.2.1. Quality assessment. The quality assessment of studies that met the inclusion/exclusion criteria was

Table 2. Total number of papers obtained through search in digital libraries

Library	Number of search results for [1]	Number of search results for [2]	Totally for [1, 2]
ACM Digital Library	68	90	158
IEEE Xplore	13	14	27
ScienceDirect	127	85	212
Springer	144	183	327
Wiley Online Library	31	58	89
MDPI	5	3	8
Total	388	433	821

conducted by the first, third, and fourth authors, while the second author reviewed the grades and calculated the average quality scores for all studies. Table 4 presents the average scores for the quality assessment of the selected studies, indicating that all studies exceeded the minimum required quality threshold for further analysis (overall quality score greater than 0.50). Each evaluator assigned a mark from the three-point scale (0.0, 0.5, or 1.0) to each study based on the quality assessment criteria. The average quality score for each study was then calculated as the mean of all average values assigned by each evaluator.

Based on the quality assessment results, the minimum score achieved was 0.53 (for study SS42), while the average quality score across all studies was 0.81.

After completing all phases of filtering the secondary studies, a total of 44 studies were selected for in-depth analysis. The selected secondary studies (SSs) are listed in Appendix A.

3.2.2. Data extraction and analysis. The data from the selected secondary studies were extracted using the template provided in Table 5. The selection of data to extract was guided by the research questions, encompassing general information about each study's publication and specific data relevant to each research

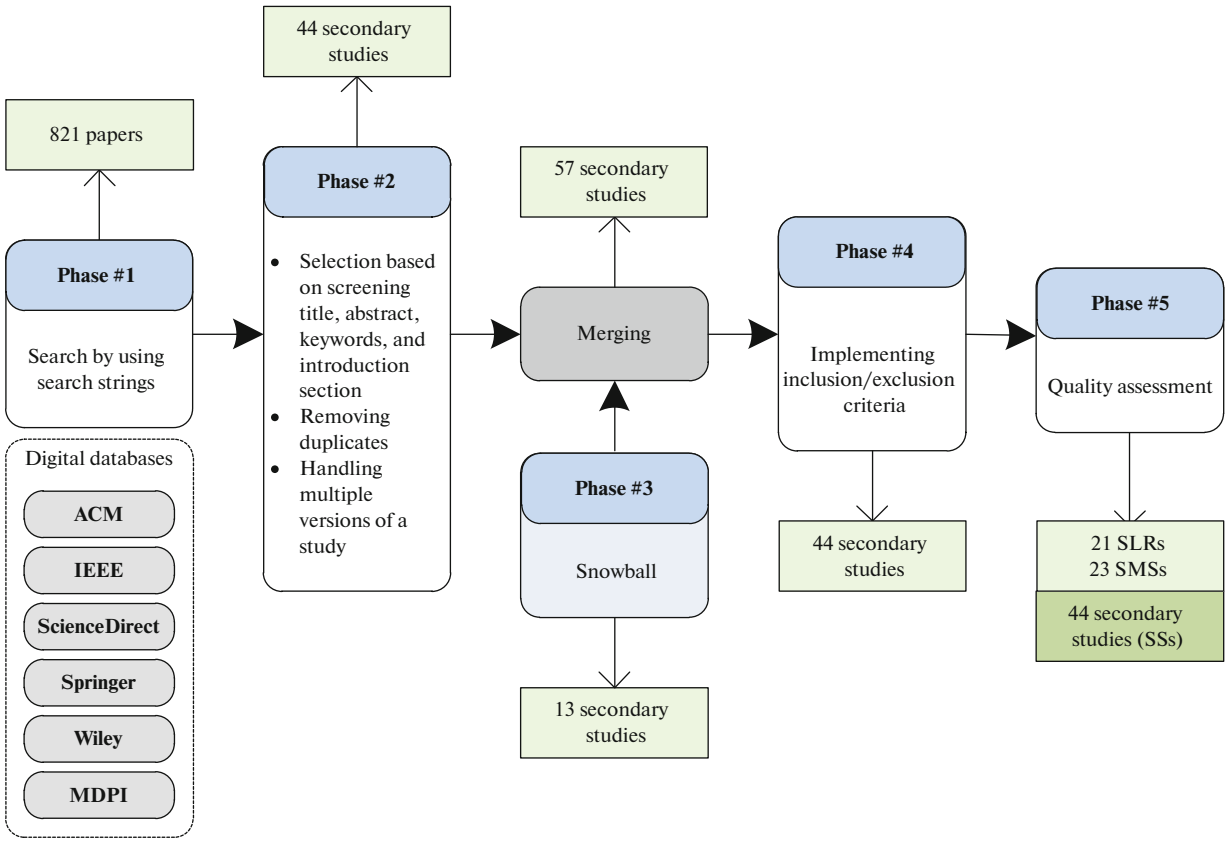


Fig. 2. The phased process of selecting secondary studies.

Table 3. Details of the phased process for selecting secondary studies

	SLRs	SMSs	MVRs	Totally SSs
[Ph#1] Selected studies after checking titles, abstracts, and keywords	19	23	14	56
[Ph#2] Selected studies after removing duplicates	16	19	9	44
[Ph#3] Snowball search for additional studies	6	7	0	13
Merging digital libraries and snowball search results	22	26	9	57
[Ph#4] Selected studies after implementing inclusion/exclusion criteria	21	23	0	44
[Ph#5] A final set of secondary studies after quality assessment	21	23	0	44

question. The extracted data was organized in an Excel spreadsheet, with each column dedicated to a specific category of extracted information.

The research findings were then synthesized using the extracted data, with the primary objective of providing comprehensive answers to the stated research questions.

4. RESEARCH FINDINGS

As mentioned earlier in the preceding section, the findings presented here are derived from the data extracted from 44 secondary studies that successfully passed all phases of the selection process. Among these studies, 23 of them are SMSs, while the remaining 21 are SLRs (refer to Table 3). The findings will be organized and presented in alignment with the research questions outlined in this study.

4.1. Publishing Trends for Secondary Studies (RQ1)

The selected secondary studies span the publication period from 2016 to 2023. Figure 3 showcases the publication trends of secondary studies based on the venue. It is obvious that the total number of studies has consistently grown over the years, and there has been a notable shift from primarily conference proceedings to journals. Out of the 44 studies, 22 were published in conference proceedings, and another 22 were published in journals. Although one study published in January 2023 does not align with this trend, it will still be utilized to address other research questions.

4.2. Quality Trends of Secondary Studies (RQ2)

Table 4 presents a detailed analysis of the quality assessment conducted on the secondary studies, revealing the following findings.

(1) The average quality score for all the selected secondary studies is 0.81. This indicates that the authors of these studies have dedicated significant attention to assessing the quality of the studies they analyzed.

(2) When comparing SLRs to SMSs, the average quality score for SLRs is 0.84, while the average quality score for SMSs is 0.77. This difference in scores is primarily attributed to the low score for Q4 in SMSs.

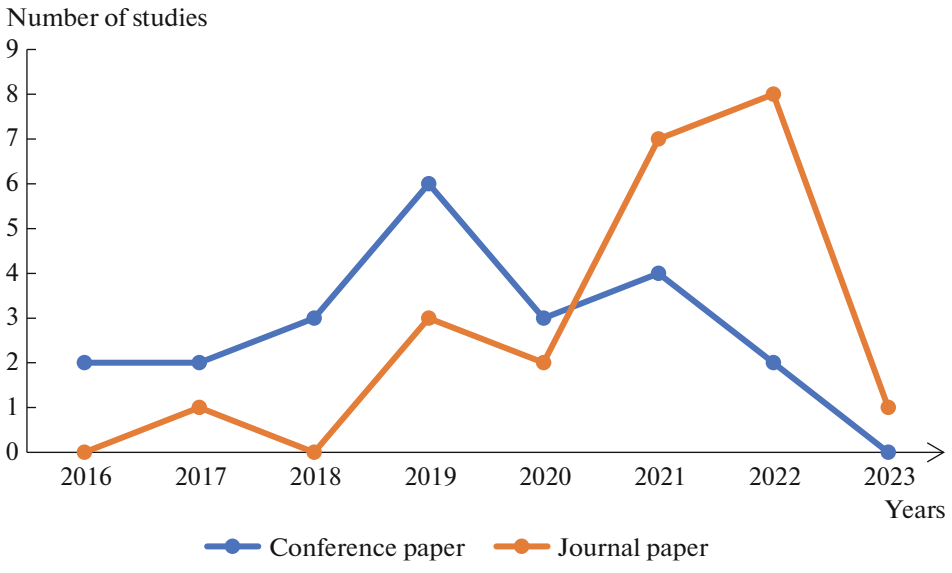


Fig. 3. Secondary studies publication trends by venue.

Table 4. Average scores for quality assessment of the selected secondary studies

ID	Year	Type	Q1	Q2	Q3	Q4	Q5	Q6	AVG
SS01	2020	SLR	0.67	0.67	0.83	0.33	0.83	0.83	0.69
SS02	2019	SLR	1.00	1.00	1.00	0.83	1.00	1.00	0.97
SS03	2019	SLR	0.83	1.00	1.00	0.83	0.83	0.83	0.89
SS04	2020	SLR	0.67	1.00	1.00	0.33	0.83	0.67	0.75
SS05	2018	SLR	0.50	0.50	0.67	0.33	0.83	0.83	0.61
SS06	2021	SLR	1.00	1.00	1.00	1.00	0.83	1.00	0.97
SS07	2021	SLR	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SS08	2018	SLR	0.50	0.67	0.67	0.50	1.00	1.00	0.72
SS09	2019	SLR	0.50	0.50	0.83	0.67	0.83	0.83	0.69
SS10	2020	SLR	1.00	1.00	1.00	0.33	1.00	1.00	0.89
SS11	2022	SLR	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SS12	2022	SLR	0.50	0.33	1.00	0.17	0.83	0.83	0.61
SS13	2022	SLR	1.00	1.00	0.83	0.00	0.83	0.83	0.75
SS14	2022	SLR	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SS15	2022	SLR	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SS16	2022	SLR	0.83	1.00	1.00	0.83	1.00	1.00	0.94
SS17	2021	SLR	1.00	1.00	0.83	0.83	1.00	1.00	0.94
SS18	2021	SLR	0.67	1.00	1.00	0.50	1.00	1.00	0.86
SS19	2018	SLR	1.00	1.00	0.67	0.83	1.00	0.83	0.89
SS20	2021	SLR	0.67	1.00	0.83	0.33	1.00	0.83	0.78
SS21	2021	SLR	0.67	1.00	1.00	0.00	0.83	0.83	0.72
SS22	2022	SMS	1.00	1.00	0.83	0.00	1.00	0.83	0.78
SS23	2017	SMS	1.00	0.83	1.00	0.00	0.67	0.83	0.72
SS24	2021	SMS	0.83	1.00	1.00	0.00	1.00	1.00	0.81
SS25	2020	SMS	1.00	1.00	1.00	0.00	0.67	1.00	0.78
SS26	2021	SMS	1.00	1.00	0.83	0.00	0.67	0.83	0.72
SS27	2017	SMS	1.00	1.00	1.00	0.00	0.67	1.00	0.78
SS28	2019	SMS	1.00	1.00	1.00	0.67	1.00	0.83	0.92
SS29	2016	SMS	1.00	1.00	0.83	0.17	1.00	1.00	0.83
SS30	2022	SMS	1.00	1.00	1.00	0.00	1.00	0.83	0.81
SS31	2022	SMS	0.83	0.83	1.00	0.75	0.50	0.83	0.79
SS32	2021	SMS	0.83	1.00	1.00	0.00	1.00	0.67	0.75
SS33	2020	SMS	1.00	0.83	1.00	0.33	1.00	1.00	0.86
SS34	2019	SMS	1.00	1.00	1.00	0.00	1.00	1.00	0.83
SS35	2021	SMS	1.00	1.00	1.00	1.00	1.00	0.83	0.97
SS36	2017	SMS	0.50	0.50	0.83	0.00	0.67	0.83	0.56
SS37	2019	SMS	1.00	1.00	1.00	0.00	0.83	0.83	0.78
SS38	2022	SMS	1.00	0.83	1.00	0.00	1.00	0.67	0.75
SS39	2021	SMS	0.83	1.00	1.00	0.00	1.00	1.00	0.81
SS40	2019	SMS	0.83	1.00	1.00	0.00	1.00	1.00	0.81
SS41	2019	SMS	0.83	1.00	1.00	0.00	0.17	1.00	0.67
SS42	2019	SMS	0.50	0.33	0.83	0.33	0.50	0.67	0.53
SS43	2016	SMS	1.00	1.00	1.00	0.17	1.00	1.00	0.86
SS44	2023	SMS	1.00	1.00	1.00	0.00	0.33	1.00	0.72
Average scores for all SSs			0.86	0.91	0.94	0.37	0.87	0.90	0.81
Average scores SLRs			0.81	0.89	0.91	0.60	0.93	0.91	0.84
Average scores SMSs			0.91	0.92	0.96	0.15	0.81	0.89	0.77

Table 5. Data extraction template

ID	Explanation	Use
D1	Study ID	Demography, RQ1
D2	Title	Demography, RQ1
D3	Year	Demography, RQ1
D4	Study type (SLR, SMS)	Demography, RQ1
D5	Venue type (conference, journal, book chapter)	Demography, RQ1
D6	Sample size (number of primary studies)	Demography, RQ1
D7	Used research methodology description	Quality assessment Q1, RQ2
D8	Inclusion/Exclusion criteria	Quality assessment Q2, RQ2
D9	Coverage of relevant studies	Quality assessment Q3, RQ2
D10	Quality assessment questions	Quality assessment Q4, RQ2
D11	Method for describing selected studies	Quality assessment Q5, RQ2
D12	Data extraction methods and tools	Quality assessment Q6, RQ2
D13	Research questions	Research topics, RQ3
D14	Research topics	Research topics, RQ3
D15	Technical implementation areas	Application area, RQ4
D16	Future research directions	Future research direction, RQ5

This distinction is influenced by the general guidelines for conducting systematic literature reviews, where quality assessment is mandatory for SLRs, while it remains optional for SMSs.

(3) The quality assessment scores for each question are presented in Table 4, revealing that there is one concerning score of 0.37 specifically related to question Q4, which pertains to the assessment of the quality of included studies. Notably, SMSs received an even lower score of 0.15 for this question. This suggests that the authors of the secondary studies did not pay enough attention to evaluating the quality of the primary studies they included, which could potentially impact the reliability of their findings.

(4) The average scores for questions Q1 (use of review methodology), Q2 (use of inclusion/exclusion criteria), Q3 (relevance of included studies), Q5 (description of included studies), and Q6 (synthesis of findings) range from 0.86 to 0.94. These results indicate that these aspects of quality assessment in the secondary studies have been addressed effectively.

Table 6 illustrates the average quality assessment scores for all the quality questions (Q1–Q6) concerning different publishing venues. Notably, studies published in journals exhibit higher score values across all the questions, suggesting that the authors have demonstrated greater diligence in assessing the quality of the primary studies due to the more rigorous peer review process associated with journal publications.

4.3. Topics Inquired in Secondary Studies (RQ3)

The identification of topic areas in the selected secondary studies is based on a thorough examination of each study, with particular attention given to the research questions and presented findings. The extracted data correspond to values in columns D13 (research questions) and D14 (research topics) in Table 5. A general overview of the topics investigated in the secondary studies is depicted in Fig. 4.

During the detailed analysis of the selected studies, 21 challenges were identified, with many of them appearing across multiple studies that focused on different topics. Subsection 4.3.5 will provide a more in-depth discussion of these challenges.

It is worth noting that while most studies have a primary focus on a specific topic, they also touch upon other related topics. For instance, it is common for a study to discuss architecture design patterns while also addressing quality attributes and specific challenges (e.g., SS27 [Research on architecting microservices: Trends, focus, and potential for industrial adoption]).

A comprehensive analysis of all the identified general topics will be presented in the subsequent subsections.

4.3.1. Architecture. Architectural design plays a vital role in the development of microservice-based software systems as it encompasses both the technical design of system functionalities and the consideration

Table 6. Average quality assessment scores regarding publication venue

	Q1	Q2	Q3	Q4	Q5	Q6	AVG
Conferences	0.83	0.89	0.91	0.29	0.83	0.88	0.77
Journals	0.90	0.92	0.97	0.44	0.91	0.92	0.84

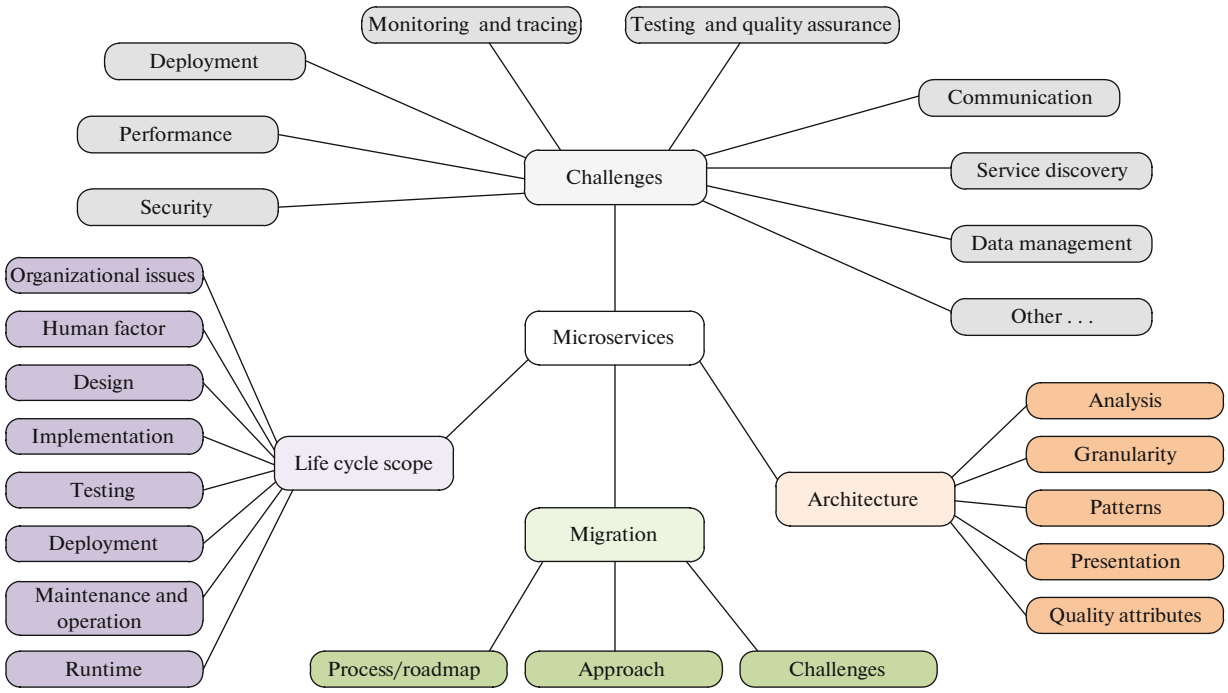


Fig. 4. General overview of microservices research topics.

of non-functional requirements, often referred to as quality attributes. Additionally, the architecture should be analyzed and presented in a manner that facilitates efficient development, operation, and maintenance of software systems. The main topics related to microservices architecture in the selected secondary studies include analysis, granularity, patterns, presentation, and quality attributes (refer to Fig. 4).

A more comprehensive examination of architecture topics, including identified subtopics and the distribution of secondary studies that mention them, is provided in Table 7.

Analysis. The analysis of software architecture is crucial for understanding both new software systems and those that require migration to microservice architectures. It focuses on the following aspects: (1) *Methods* [SS09][SS27][SS35][SS39][SS44]: Various methods can be distinguished, including static analysis, dynamic analysis, combined dynamic and static analysis, model-based analysis, graph-based analysis, and pattern-based analysis; (2) *Tools* [SS30][SS39]: Specific tools tailored to each analysis method, which facilitate more efficient and automated work; and (3) *Challenges* [SS30][SS39]: These encompass architectural analysis, migration to microservices, software architecture reconstruction, technical debt analysis, microservice evolution, quality attribute analysis, and fault analysis. By considering these aspects, a comprehensive understanding of software architectures, especially in the context of microservices, can be achieved.

Granularity. Granularity refers to the size of individual microservices within a software system based on the microservices architecture. It plays a crucial role in determining the functioning of the system and its quality attributes, including performance, maintainability, data storage, and scalability. The granularity of microservices has a direct impact on how well the system performs and how effectively it can be maintained and scaled to meet changing demands. Consideration of granularity is essential when developing a new system or migrating legacy systems to a microservices architecture [SS09][SS24]. Determining the optimal granularity involves finding the right balance between the level of functionality encapsulated within each microservice and the need for modularity, maintainability, and scalability. By carefully assessing the purpose and requirements of the system, developers can determine the appropriate size and scope of each microservice to achieve the desired architectural goals. Insights from the selected studies on granularity reveal the following main subtopics: (1) *Methods or approaches for defining granularity* [SS04][SS09][SS17][SS24]; (2) *Metrics used for evaluating granularity* [SS17]; and (3) *Quality attributes affected by granularity* [SS17]. In a detailed inquiry presented in [SS17], various approaches are used to define the granularity of microservices. These include the trial and error approach, the approach based on the number of lines of code, the approach related to implementation units, the approach related to business capabilities, and the approach using domain-driven design. Additionally, findings from [SS09] clas-

Table 7. Topics and subtopics related to a microservices architecture

Topic	Subtopic	Secondary studies
Analysis	Methods	SS09, SS27, SS35, SS39, SS44
	Tools	SS30, SS39
	Challenges	SS30, SS39
Granularity	Approach	SS04, SS09, SS17, SS24
	Quality attributes	SS17
	Metrics	SS17
Patterns	Design	SS03, SS05, SS19, SS27,SS33, SS34, SS37, SS43
	Composition	SS02, SS09, SS19, SS33, SS43
	Communication	SS03, SS05, SS06, SS19, SS33, SS37
	Deployment	SS03, SS06, SS37, SS19, SS33, SS37
	Data storage	SS03, SS19, SS33, SS37
	Antipattern	SS01, SS44
Presentation	Languages	SS22, SS27, SS29, SS33
	Diagrams	SS27, SS29
	Visualization	SS30
Quality attributes	Reliability	SS03, SS08, SS17, SS27, SS33, SS34
	Security	SS03, SS07, SS08, SS17, SS27, SS29, SS33, SS40
	Compatibility	SS03, SS27, SS33, SS34, SS40
	Maintainability	SS03, SS08, SS17, SS27, SS29, SS33, SS34
	Performance	SS03, SS07, SS17, SS19, SS27, SS29, SS33, SS34, SS40
	Portability	SS03, SS27, SS33, SS34
	Testability	SS07, SS33
	Availability	SS07, SS08, SS17, SS33
	Monitorability	SS07, SS33
	Scalability	SS07, SS08, SS17, SS19, SS27, SS29, SS33, SS34, SS40
	Modularity	SS17, SS29
	Other...	SS29, SS33, SS40

sify approaches for determining granularity as static code analysis aided, meta-data aided, workload-data aided, and dynamic microservice composition, all considering different levels of granularity within software systems. Common metrics used for measuring granularity encompass coupling, cohesion, complexity, performance, quality attributes, use of computational resources, development team, and source code. Regarding quality attributes, granularity has an impact on runtime attributes such as scalability, performance, reliability, availability, and functionality, as well as attributes related to software as an artifact, such as maintainability, modularity, and reusability.

Patterns. The identification and categorization of specific challenges and their corresponding solutions during software development, operation, and maintenance contribute to the recognition of recurring scenarios that can be addressed with specific solutions, commonly referred to as patterns. Incorporating patterns into the software life cycle enables developers to find reliable solutions to common problems, enhances communication among team members and with clients, and aids in meeting quality requirements. The analysis of the secondary studies reveals that a significant portion of them directly or indirectly explore patterns associated with different facets of microservices architecture. Additionally, some studies delve into adapting patterns in other domains, such as DevOps or their application in IoT-based smart systems. The secondary studies identify the following categories of patterns: (1) *Design* [SS03][SS05][SS19][SS27][SS33][SS34][SS37][SS43]—patterns used for structuring and organizing microservices efficiently. Commonly used patterns include API gateway, publish/subscribe, circuit breaker, proxy, and load balancer [SS27]); (2) *Composition* [SS02][SS09][SS19][SS33][SS43]—pat-

terns related to composing different microservices in a software system. Commonly used patterns include semantic annotation [SS02], best-fitting, and workload-based approaches [SS09]); (3) *Communication* [SS03][SS05][SS06][SS19][SS33][SS37]—patterns related to communication between multiple microservices. Commonly used patterns include synchronous communication, publish/subscribe communication, combination of HTTP and message queue, communication using message-oriented middleware, asynchronous communication, point-to-point communication, and communication using binary protocols [SS06]); (4) *Deployment* [SS03][SS06][SS37][SS19] [SS33][SS37]—patterns related to the deployment or distribution of microservices to multiple resources for operational use. Commonly used patterns include serverless deployment, service instances per VM, and service instances per container [SS06]); and (5) *Data storage* [SS03][SS19][SS33][SS37]—patterns oriented towards improving performances of data management systems. Commonly used patterns include the database-per-service pattern, the database cluster pattern, and the shared database server pattern [SS37]).

In addition, the authors discussed *antipatterns* in [SS01] and [SS44], which refer to design and implementation choices that result in inadequate/poor software system design, leading to issues during operation and maintenance. In [SS01], antipatterns are categorized in a catalog that covers the design, implementation, deployment, and monitoring of microservices. On the other hand, [SS44] focuses on antipatterns and specific software tools for visualizing microservices during runtime.

Presentation. Efficient design and utilization of microservice-based systems require the presentation of software architecture through various approaches that depict the system's structure and behavior. It is crucial to consider the dynamic characteristics of the system during both the development phase and operational use, which necessitates effective visualization. The secondary studies identified the following subtopics: (1) *Languages* [SS22][SS27][SS29][SS33]—This pertains to the use of specific languages for describing the architecture, such as RAML, YAML, Jolie, or various pseudocodes [SS27][SS29]; (2) *Diagrams* [SS27][SS29]—this involves the utilization of visual diagrams, ranging from informal drawings to specialized and sophisticated diagrams such as Component/Container, Process/Behavior, Sequence, Execution Timeline, Deployment, Class, Use Case, Type Graph, Instance Graph, and Dependency Graph [SS29]; and (3) *Visualization* [S30]—this subtopic focuses on the application of dynamic analysis techniques, supported with specialized tools, for analyzing or recovering software architecture. Techniques such as process mining, distributed tracing, and monitoring are employed for this purpose.

Quality attributes. Quality characteristics of software systems are typically reflected in the form of quality attributes, which are associated with non-functional requirements. While there are numerous quality attributes, it is often challenging to satisfy all of them simultaneously for a specific system. In such cases, trade-offs must be made when considering quality attributes. Table 7 presents the most common quality attributes mentioned in many studies, including *Reliability*, *Security*, *Compatibility*, *Maintainability*, *Performance*, *Portability*, *Testability*, *Availability*, *Monitorability*, *Scalability*, and *Modularity*. These attributes have been extensively discussed in various sources and are relevant throughout the software life cycle, from design to operational use and maintenance. The selected secondary studies also mention additional quality attributes, such as *Modifiability* [SS33], *Usability* [SS33], *Deployability* [SS33], *Flexibility* [SS33], *Reusability* [SS29], *Manageability* [SS29], *Independence* [SS29], *Traceability* [SS40], *Complexity* [SS40], *Load balancing* [SS29], and *Organizational alignment* [SS29].

To effectively address quality attributes, appropriate metrics need to be proposed, such as time, complexity, number of requests, or number of affected files [SS03][SS17]. These metrics facilitate continuous improvements in quality attributes over time [SS07].

4.3.2. Life cycle scope. Microservice-based systems, like other software systems, undergo various lifecycle phases that bring forth unique challenges, necessitating the utilization of specific methods, approaches, and tools. It is crucial to consider not only technical and technological aspects but also human and organizational factors that impact activities throughout the entire lifecycle. The primary topics identified in the selected secondary studies pertaining to the life cycle scope of microservices encompass *Design*, *Implementation*, *Testing*, *Deployment*, *Maintenance and Operation*, *Runtime*, *Organizational issues*, and the *Human factor* (refer to Fig. 4).

A more comprehensive review of life cycle scope topics, along with their respective subtopics and the distribution of secondary studies mentioning them is given in Table 8.

Design. The design phase of the lifecycle is crucial for achieving the desired system structure and fulfilling the proposed quality characteristics. Topics in this phase are closely related to design patterns, architecture representation, and quality attributes, all of which fall under the general topic of Architecture. The following subtopics have been identified in the secondary studies: (1) *Identification strategies* [SS02][SS04]— this subtopic focuses on the identification of services during the design of complex systems; (2) *Domain-Driven Design* [SS04][SS39]—this subtopic relates to the use of principles, patterns, and domain-specific knowledge during system design; (3) *Representations* [SS04][SS34][SS36][SS39]—this subtopic is about the

Table 8. Topics and subtopics related to the life cycle scope of microservices

Topic	Subtopic	Secondary studies
Design	Identification strategies	SS02, SS04
	Domain driven design	SS04, SS39
	Representations	SS04, SS34, SS36, SS39
	Design for failure	SS08, SS36
Implementation	Technology stack	SS08, SS36, SS37
	Supporting systems	SS08, SS34, SS36, SS37
	Services interfaces	SS08, SS34
Testing	Approaches	SS14, SS23, SS25, SS36, SS37, SS40
	Tools	SS25, SS37, SS40
Deployment	Platform	SS08, SS32, SS34, SS37
	Monitoring	SS08, SS14, SS23, SS32, SS34, SS36, SS39
	Approaches	SS06, SS23, SS32, SS37
Maintenance & Operation	Load balancing	SS11, SS34
	Fault diagnosis	SS11, SS23, SS36, SS37, SS39
	Autoscaling	SS11, SS12, SS21, SS34, SS37
	Anomaly detection	SS13, SS37
	Resource scheduling	SS13, SS33
	Analysis	SS36, SS39
Runtime	Virtualization	SS08, SS43
	Discovery	SS14, SS43
	Control	SS08, SS34, SS39
	Verification and validation	SS08, SS21
	Visualization	SS30, SS39, SS44
Organizational issue	DevOps	SS08, SS19, SS25, SS32, SS33, SS34, SS37, SS43
	Continuous delivery–integration–deployment	SS08, SS25, SS43
Human factor	Roles	SS04, SS17
	Skills	SS04, SS38

use of various methods and tools for representing microservice system being developed; and (4) *Design for failure* [SS08][SS36]—this subtopic is related to design principles and methods that enable the design of systems with increased fault tolerance, self-healing capabilities, and variability characteristics. These subtopics provide valuable insights into the design phase and its importance in creating robust microservice-based systems.

Implementation. The implementation phase utilizes the products and decisions from the design phase to create microservices and integrate them into a cohesive system. Through an analysis of the secondary studies, the following subtopics have been identified: (1) *Technology stack* [SS08][SS36][SS37]—this subtopic pertains to the use of various languages (formal, scripting, object-oriented), interaction models for communication flow, and protocols for data exchange (e.g., REST/HTTP, RPC-alike, message queues); (2) *Supporting systems* [SS08][SS34][SS36][SS37]—this subtopic focuses on data storage systems for distributed microservices (e.g., SQL, graph-oriented, document-oriented) and systems for service discovery in a dynamic environment with heterogeneous microservices (e.g., client-drive, server-driven); and (3) *Service interfaces* [SS08][SS34]—this subtopic concerns the specification of contracts for microservices communication, encompassing formal, tech-tied, and ad-hoc definitions in a novel language. These subtopics shed light on the implementation phase and the important considerations related to technology selection, supporting systems, and service interfaces in the development of microservice-based systems.

Testing. The complex nature and dynamic behavior of microservice-based systems present several challenges in their testing, as it requires a deep understanding of the intricate and concurrent behaviors of microservices and their mutual interactions. Through an

analysis of the secondary studies, the following subtopics have been identified: (1) *Approaches* [SS14][SS23][SS25][SS36][SS37][SS40]—this subtopic encompasses various testing approaches employed during development, ranging from unit testing to integration testing., It also includes continuous testing as part of DevOps and continuous engineering practices, testing of microservices and system performance, testing during migration, and model-based testing; and (2) *Tools* [SS25][SS37][SS40]—this subtopic pertains to the tools utilized in the testing process, with a preference for automated testing. It includes libraries and frameworks that enable specific types of tests. These subtopics shed light on the challenges associated with testing microservice-based systems and highlight the importance of employing appropriate approaches and tools for comprehensive and effective testing throughout the software development lifecycle.

Deployment. A deployment practice encompasses activities, methods, and tools necessary for the establishment of heterogeneous microservices to meet the requirements of contemporary businesses. Automated and continuous development and deployment processes are essential in ensuring the reliable and scalable delivery and operation of microservice-based systems. The analysis of the secondary studies has identified the following subtopics: (1) *Platform* [SS08][SS32][SS34][SS37]—this subtopic pertains to the selection of a hosting system for running microservices, such as public cloud, private cloud, or in-house infrastructure; (2) *Monitoring* [SS08][SS14][SS23][SS32][SS34][SS36][SS39]—this subtopic relates to the activities performed to prevent or respond to failures or changes in the environment. It encompasses the use of built-in cloud services, third-party services, and ad-hoc solutions for monitoring microservices; and (3) *Approaches* [SS06][SS23][SS32][SS37]—this subtopic addresses the various ways and strategies employed to facilitate the utilization of microservices-based systems. It includes considerations such as hosting multiple services per host or adopting a single service-per-host approach. These subtopics highlight the importance of selecting the appropriate platform, implementing robust monitoring practices, and employing effective approaches for deploying and utilizing microservice-based systems.

Maintenance and operation. The primary focus of maintenance and operation activities in the software life cycle is to ensure the usability and operability of the software. Effective maintenance activities are crucial for ensuring the stable and reliable functioning of complex microservice-based software systems. The following subtopics have been identified: (1) *Load balancing* [SS11][SS34]—this subtopic pertains to the coordination and management of a large number of service requests in systems with heterogeneous and distributed microservices. It involves the use of algorithms and performance measurement techniques to optimize load distribution; (2) *Fault diagnosis* [SS11][SS23][SS36][SS37][SS39]—this subtopic involves improving the quality and efficiency of software operation by detecting faults. It includes activities such as monitoring and localization of faults, identifying fault types, and fault modeling; (3) *Autoscaling* [SS11][SS12][SS21][SS34][SS37]—autoscaling relates to the adjustment of system resources to meet changing needs and growing requirements. This subtopic covers resource prediction and scheduling methods, as well as resource allocation and recovery techniques; (4) *Anomaly detection* [SS13][SS37] – Anomaly detection focuses on identifying critical behaviors or abnormal states in application performance. It involves techniques for detecting and handling anomalies to ensure system stability; (5) *Resource scheduling* [SS13][SS33]—this subtopic involves the dynamic adjustments of system resources in response to the overall system state and workload. It includes techniques for efficiently managing and allocating resources based on demand; and (6) *Analysis* [SS36][SS39]—this subtopic covers methods and tools for analyzing the states and behaviors of microservice-based systems during operational use. It includes static analysis, dynamic analysis, combined dynamic and static analysis, model-based analysis, graph-based analysis, and pattern-based analysis. All these subtopics highlight the importance of load balancing, fault diagnosis, autoscaling, anomaly detection, resource scheduling, and analysis in the maintenance and operation of microservice-based systems.

Runtime. Analyzing the architecture, functioning, and performance of microservices-based systems requires extracting information from both static and dynamic sources during runtime. Monitoring the dynamics of the system is crucial because changes in structure and communication often occur during system execution. The following subtopics have been identified: (1) *Virtualization* [SS08][SS43]—this subtopic pertains to different levels of platform abstraction, isolation, and sharing. It includes technologies such as virtual machines, containers, serverless computing, and non-virtualized environments; (2) *Discovery* [SS14][SS43]—discovery relates to identifying and finding appropriate services based on workload, scalability, and service quality considerations. This subtopic covers mechanisms and optimization techniques for efficient service discovery; (3) *Control* [SS08][SS34][SS39]—control involves managing execution at both the local level of individual microservices and the system level as a whole. It includes activities such as monitoring, planning, execution, and evaluation to ensure effective control over the system; (4) *Verification and Validation* [SS08][SS21]—this subtopic focuses on assessing the quality of microservices during runtime. It includes the use of models and properties for verification and validation purposes; and (5) *Visualization* [SS30][SS39][SS44]—visualization relates to visually representing the microservices architecture during runtime. This subtopic covers

techniques, tools, and types of information that can be extracted and displayed for better understanding and analysis. Overall, these subtopics highlight the importance of virtualization, discovery, control, verification and validation, and visualization in the analysis of microservices-based systems.

Organizational issue. Software development, operation, and maintenance take place within specific organizational contexts, including software development organizations and client organizations. These contexts have their own processes, procedures, challenges, and cultures that impact the technical and technological aspects of software processes. The following subtopics have been identified: (1) *DevOps* [SS08][SS19][SS25][SS32][SS33][SS34][SS37][SS43]—DevOps encompasses the cultural and practical aspects of organizing stakeholders involved in the development and operation of microservices-based systems. It involves addressing people-related issues, establishing effective processes, and utilizing appropriate tools and technologies; and (2) *Continuous Delivery, Integration, and Deployment* [SS08][SS25] [SS43]—this subtopic focuses on continuous activities that facilitate a seamless transition between phases in microservices-based systems. It includes practices related to continuous delivery, integration, and deployment to ensure smooth and efficient software development and deployment processes. These subtopics highlight the importance of considering organizational contexts, including DevOps practices and continuous delivery, integration, and deployment, to effectively address the challenges and complexities associated with software development, operation, and maintenance within different organizations.

Human factor. Even though the literature primarily emphasizes technical and technological aspects, it is important to recognize that all activities in the software systems' life cycle are ultimately carried out and supervised by people. The following subtopics have been identified: (1) *Roles* [SS04][SS17]—this subtopic pertains to the various roles that individuals assume in the microservice life cycle within software organizations and client organizations; and (2) *Skills* [SS04][SS38]—this subtopic focuses on the technical and soft skills that are necessary for individuals in different roles within the microservices' life cycle. It encompasses the specific knowledge, expertise, and abilities required to effectively perform their respective tasks. These subtopics underscore the significance of considering the human factor in addition to the technical and technological aspects of software development. By understanding the roles and skills of individuals involved in the microservice life cycle, organizations can better support their teams and ensure the successful execution of software projects.

4.3.3. Migration. One of the primary challenges with legacy software systems, built on monolithic architecture, is the need for subsequent modifications to keep them operational and useful for end users. However, frequent modifications can increase software complexity, reduce performance, and make maintenance challenging. One common solution to address these issues is migrating software systems to microservices architectures. In the selected secondary studies, the main topics related to migration to microservices are *Approach*, *Process/roadmap*, and *Challenges* (refer to Fig. 4). A more detailed review of migration topics, including identified subtopics and the distribution of secondary studies that mention them, is presented in Table 9.

Approach. In practice, different types of legacy systems require varying approaches for modernization and migration to microservice architectures. These approaches necessitate the use of specific strategies, methods, and tools while also presenting specific challenges. The identified subtopics in the secondary studies are as follows: (1) *Strategy* [SS24][SS13][SS26][SS38]—this subtopic pertains to the overall strategy chosen in a migration project, such as clustering, candidate identification based on quality attributes, data-driven approaches, or bottom-up approaches; (2) *Decomposition method* [SS09][SS24][SS13][SS26]—this subtopic focuses on the selection of the analysis method used to decompose the legacy system and identify microservices. Examples include Static Code Analysis, Meta-Data aided, Workload-Data aided, or Dynamic Microservice Composition; (3) *Unit level* [SS09][SS24][SS38]—this subtopic involves selecting the most suitable level of software artifacts during the decomposition of the old system and migration to microservices. Options may include business functions, database tables, classes, use cases, or EJBs; and (4) *Evolution* [SS09][SS24][SS38]—this subtopic relates to supporting the scalability and maintainability features of the new microservice-based system throughout the migration process. All these subtopics highlight the diverse considerations and decisions involved in modernizing and migrating legacy systems to microservices. They provide insights into the strategies, decomposition methods, unit levels, and evolutionary aspects that are crucial for successful migration projects.

Process/roadmap. Every migration project follows a process or roadmap that guides the organization of activities and determines the input and output information. Additionally, specific factors can significantly influence the migration process, depending on the unique characteristics of each project. The identified subtopics in the secondary studies are as follows: (1) *Definition* [SS24][SS26][SS38]—this subtopic pertains to the selection or proposal of different processes, guidelines, and roadmaps, supported by specific tools, to facilitate the migration process; (2) *Input information* [SS24][SS26][SS38]—this subtopic focuses on identifying the required input information for the migration process, such as documents, APIs, non-functional requirements, tools, and the team's exper-

Table 9. Topics and subtopics related to migration to microservices

Topic	Subtopic	Secondary studies
Approach	Strategy	SS24, SS13, SS26, SS38
	Decomposition method	SS09, SS24, SS13, SS26
	Unit level	SS09, SS24, SS38
	Evolution	SS09, SS24, SS38
Process/roadmap	Definition	SS24, SS26, SS38
	Input information	SS24, SS26, SS38
	Output information	SS24, SS26
	Success factors	SS38
	Motivation	SS24
Challenges	Technical	SS24, SS26, SS38
	Organizational	SS26, SS38
	Knowledge and skills	SS04, SS26, SS38

tise; (3) *Output information* [SS24][SS26]—this subtopic relates to the information produced upon completing the migration process, including microservices candidates, communication approaches, and system architecture; (4) *Success factors* [SS38]—this subtopic encompasses the factors that influence the successful execution and completion of the migration process, such as system architecture, time and cost considerations, replacement strategies, achievable attributes, application size, and complexity; and (5) *Motivation* [SS24]—this subtopic explores the motivations or driving factors behind organizing a migration project, which can be technical, operational, or organizational in nature. All these subtopics shed light on the various elements involved in migration projects, ranging from defining processes and gathering input information to generating output information, considering success factors, and understanding the motivations behind the project. They provide insights into the comprehensive framework necessary for a successful and well-organized migration process.

Challenges. The migration of existing systems to a new microservice-based architecture is a challenging project that poses various obstacles for organizations, teams, and individuals involved. The identified subtopics in the secondary studies are as follows: (1) *Technical* [SS24][SS26][SS38]—this subtopic focuses on the introduction of new technologies and the selection of the most suitable tools for the migration process; (2) *Organizational* [SS26][SS38]—this subtopic pertains to organizational changes within an IT company that undertakes a migration process. It involves reorganizing the hierarchical structure of the company and establishing independent teams to work on individual services, promoting a more efficient and effective workflow.; and (3) *Knowledge and skills* [SS04, SS26, SS38]—this subtopic emphasizes the importance of selecting team members with the appropriate knowledge and a combination of technical and non-technical skills necessary for the successful implementation of the migration project. It highlights the significance of expertise and competency within the team. These subtopics highlight the significant aspects that need to be addressed during the migration of existing systems to a microservice-based architecture. The technical considerations, organizational adjustments, and the importance of assembling a knowledgeable and skilled team contribute to the overall success of the migration project.

4.3.4. Challenges. The adoption of microservices in industrial practice presents numerous challenges for both practitioners and researchers, given the inherent complexity and heterogeneity of microservice-based systems. Exploring and addressing the problems and challenges identified in the literature can lead to solutions that enhance the performance of software systems in practical settings. Figure 4 illustrates eight challenges that have been mentioned in at least three secondary studies, with additional challenges categorized under the "Other ..." section. A comprehensive list of 21 identified challenges, along with the corresponding secondary studies that reference them, is presented in Table 10. By understanding and tackling these challenges, practitioners, and researchers can make significant strides in overcoming the obstacles inherent in microservice adoption and improve the effectiveness of software systems.

Upon reviewing the secondary studies, it becomes evident that regardless of the topics covered, all studies identified specific challenges and proposed corresponding solutions. Among the challenges identified, *Security* emerges as the most crucial, with 13 occurrences across the studies. These discussions encompass a wide range of topics, needs, and scenarios [SS16], with proposed taxonomies or frameworks to address security issues [SS14]. Following closely, *Communication* is mentioned in seven studies as the second most frequently cited challenge. Communica-

Table 10. Challenges related to microservices

Challenge	Secondary studies	Frequency
Security	SS14, SS15, SS16, S18, SS23, SS27, SS28, SS29, SS33, SS34, SS35, SS36, SS42	13
Communication	SS06, SS14, SS18, SS23, SS27, SS29, SS34	7
Testing and quality assurance	SS14, SS23, SS25, SS27, SS34, SS36	6
Performance	SS14, SS23, SS29, SS33, SS36	5
Deployment	SS06, SS23, SS29, SS32, SS36	5
Monitoring and tracing	SS14, SS23, SS29, SS32, SS36	5
Service discovery	SS14, SS23, SS29	3
Data management	SS14, SS27, SS34	3
Scalability	SS21, SS36	2
Migration	SS26, SS38	2
Complexity	SS27, SS34	2
Composition	SS27, SS34	2
Decomposition	SS14	1
Orchestration	SS14	1
Modeling	SS23	1
Context awareness	SS23	1
Integration	SS29	1
Fault tolerance	SS29	1
Publishing	SS32	1
Upgrading	SS32	1
Availability	SS36	1

tion challenges may arise from remote calls, during replication of services or data, or service discovery [SS06]. The third most frequently mentioned challenge pertains to *Testing and quality assurance* of microservice-based systems. This encompasses various specific challenges, including faster test feedback, automated testing, intercommunication testing, granularity testing, runtime testing, integration testing, and performance testing [SS25].

Deployment poses a significant challenge due to the complexity involved in deploying heterogeneous microservices. This challenge encompasses deployment coordination, cost considerations, required configurations, continuous integration, continuous delivery, and managing independent microservice failures [SS06]. Another important challenge pertains to the *Performance* quality attribute, which receives the most attention, especially when trade-offs between multiple attributes are necessary [SS07]. *Monitoring and tracing* microservice-based systems involve monitoring runtime performance, overseeing the infrastructure supporting microservices execution, and logging data for further analysis to enhance system performance [SS30]. *Data management* presents challenges related to maintaining service independence, ensuring data consistency, and governing distributed transactions [SS14]. *Service discovery* poses a challenge during system design and implementation, where the selection of appropriate discovery mechanisms is crucial. Additionally, during runtime, it becomes essential to discover services that meet the required quality parameters [SS14].

Overall, all the identified challenges, as presented in Table 10, require careful consideration in specific design or usage scenarios. This involves exploring potential solutions for identified problems and making informed decisions to ensure the desired fulfillment of system quality attributes.

4.4. Technical Implementation and Integration (RQ4)

The dynamics and the increasing need for integrating heterogeneous and complex systems in various sectors such as healthcare, industry, and transportation necessitate the adoption of service-oriented architectures to facilitate distributed processing capabilities and data integration. Microservices have emerged as a popular solution due to their widely recognized advantages, leading to their widespread use across different business domains and in various aspects of human life. The selected secondary studies highlight three main topics related to the technical implementation and integration, including *Service type*, *Service domain*, and *Industry adoption* (refer to Fig. 4). For a more comprehensive examination of technical implementation and integration, Table 11

Table 11. Topics and subtopics related to technical implementation and integration of microservices

Topic	Subtopic	Secondary studies
Service type	Functional	SS05, SS34
	Infrastructure	SS05, SS34
Domain	Smart systems	SS05, SS09, SS31, SS43 (smart city, IoT)
	Fog applications	SS42
	Big data	SS20
	Blockchain	SS31
	Enterprise	SS31, SS33, SS43
Industry adoption	Readiness level	SS27, SS34
	Industry involvement	SS27, SS34
	Tools and system support	SS27, SS34
	Evaluation and benchmarking	SS27, SS31, SS34

provides a detailed overview of the identified related subtopics and the distribution of secondary studies that mention them.

4.4.1. Service type. Microservices are deployed within complex and heterogeneous systems, requiring sophisticated infrastructures for hosting and execution. Apart from the functional services that cater to user needs, it is crucial to deploy services that support the infrastructure essential for the proper functioning of microservices. The secondary studies have identified two main subtopics in this context: (1) *Functional services* [SS05][SS34]—these services are responsible for providing functionalities to users, enabling them to perform tasks or access features; and (2) *Infrastructure services* [SS05][SS34]—these services encompass non-functional requirements, infrastructure, and service monitoring, as well as maintenance tasks that are not directly related to user functionalities but are crucial for the overall functioning of the system. By understanding and addressing these subtopics, organizations can ensure the efficient deployment and management of microservices within their infrastructures.

4.4.2. Domain. The secondary studies have identified the following subtopics: (1) *Smart systems* [SS05][SS09][SS31][SS43]—this subtopic focuses on the implementation of microservices for technical integration in smart systems, such as smart cities, smart transportation, and IoT applications; (2) *Fog applications* [SS42]—this subtopic explores the use of microservices in fog computing applications, including motivations, challenges, and methods; (3) *Big data* [SS20]—this subtopic examines the utilization of microservices in Big Data applications, exploring the interplay between microservices and Big data, as well as relevant methods and application domains; (4) *Blockchain* [SS31]—this subtopic investigates the combination of microservices with blockchain technologies, addressing methods, challenges, and domains where these technologies are applied together; and (5) *Enterprise* [SS31][SS33][SS43]—this subtopic focuses on the use of microservices in various business domains, such as healthcare, online commerce, supply chain management, financial systems, and telecommunications. All these subtopics provide insights into the specific application areas where microservices are utilized, enabling researchers and practitioners to explore and contribute to these domains.

4.4.3. Industry adoption. The majority of studies report the widespread acceptance and adoption of microservices in various business scenarios, establishing them as a prominent software development approach in the software and IT industry. However, adopting microservices introduces several issues that warrant investigation to enhance practices in the software industry. The identified subtopics in the secondary studies are as follows: (1) *Readiness level* [SS27][SS34]—this topic pertains to the maturity level of specific methods, tools, and technologies intended for implementation in industrial projects. It focuses on assessing the readiness of these resources for practical adoption; (2) *Industry involvement* [SS27][SS34]—this subtopic explores the degree of engagement and participation of industry experts in research projects. Collaboration with industry professionals enhances knowledge sharing and promotes practical applicability; (3) *Tools and system support* [SS27][SS34]—this subtopic examines the utilization of specialized tools and systems that support the development and operation of microservice-based systems. Examples include knowledge-based tools and open-source test systems that facilitate the implementation and management of such architectures; and (4) *Evaluation and benchmarking* [SS27][SS31, SS34]—this subtopic emphasizes the use of evaluation and benchmarking tools to assess the quality and effectiveness of designed microservice-based architectures. Evaluation and benchmarking provide valuable insights into the performance and viability of these architectures. The aforementioned subtopics shed light on critical aspects related to microservices adoption, including readiness, industry

collaboration, tooling support, and evaluation practices. Exploring and addressing these areas can contribute to further improvements in the software industry.

4.5. Directions for Future Research (RQ5)

When considering the enumeration and listing of potential future research directions, it should be noted that these directions have been identified by analyzing the discussion sections and concluding remarks of each examined secondary study. These sections also highlight the emerging open challenges that need to be addressed. Therefore, we searched within the secondary studies using keywords such as "future" in combination with the words "research," "work," and "direction," as well as the keywords "further," "challenge," and "gap."

Notably, the majority of authors indicate future research directions in two main ways: (a) by highlighting and elaborating on potential research gaps, open challenges, and issues encountered during their analysis, and using them as a basis to propose broader research directions through calls for action to the community of researchers and practitioners in the field of microservice architectures; and (b) by proposing specific research directions that are closely related to the research methodology employed in their study. These specific directions aim to promote the effective development and application of microservice architectures.

In the coming years, the research focus on architecting with microservices, which is expected to shape their future, will center on addressing the following issues.

• Exploring various types of microservice architectures, their structure, and design aspects. This involves expanding the existing knowledge base in a systematic, structured, and consistent manner, (a) by *including both theoretical and practical learning and exploration* [SS07][SS10][SS19][SS23][SS26][SS29][SS32][SS37][SS39][SS41][SS43], (b) by *investigating the application of microservices in specific domains* [SS20] or *real-world scenarios* [SS21][SS24][SS25][SS27][SS30][SS31][SS33], and (c) *by considering organizational factors* [SS26][SS38] and *addressing human-related issues* [SS27][SS31][SS32][SS37][SS38] in microservice architectures. These research directions aim to enhance our understanding of microservices and contribute to the development of a comprehensive and practical framework for their successful implementation.

• To enhance the current state and facilitate the development and operation of future microservice architectures, *various novel and innovative methods* [SS16][SS17][SS26][SS39], *models* [SS13][SS14][SS39][SS41], *frameworks* [SS18][SS26][SS44], *infrastructures* [SS06], *techniques* [SS11][SS17][SS28], *algorithms* [SS39], *migration strategies* [SS16][SS24], *approaches* [SS06][SS09][SS25][SS29], *technologies* [SS31][SS34], *design patterns* [SS03][SS15][SS44] and *antipatterns* [SS01][SS44], *languages* [SS22], and *tools* [SS04][SS16][SS22][SS24][SS25][SS26][SS30][SS39][SS44] can be utilized. These resources offer opportunities for advancements in microservice architectures, enabling more efficient and effective development and operation [SS08][SS14][SS32]. By leveraging these innovative elements, researchers and practitioners can explore new possibilities and overcome existing challenges in this domain.

• To ensure the quality of microservice architectures, it is crucial to assess the associated quality attributes [SS19]. These attributes encompass *scalability* [SS12][SS21][SS22], *performance* [SS24][SS33][SS43], *availability* [SS11], *monitorability* [SS33], *reliability* [SS11][SS25][SS40], *security* [SS16][SS27][SS28][SS33][SS34], *testability* [SS25][SS27][SS34], *maintainability* [SS11][SS43], *resilience* [SS25], *interoperability* [SS40], and *portability* [SS21][SS27][SS34]. It is also important to consider their *complex interactions and trade-offs* [SS03][SS07][SS25][SS34][SS39][SS42][SS43]. By evaluating and addressing these attributes, researchers and practitioners can optimize the performance, robustness, and overall effectiveness of microservice architectures. This comprehensive assessment can help identify areas for improvement and guides the development of resilient and adaptable systems.

• Focusing on the industrial adoption of microservice architectural designs, which involves several crucial phases, including *implementation*, *validation* [SS02][SS16][SS27], *operation*, *deployment* [SS06], *maintenance*, and *testing* of microservice architectural designs in practice [SS05][SS10][SS11][SS25][SS26][SS32][SS34][SS35][SS36][SS37][SS41]. These phases play a significant role in ensuring the successful implementation and long-term viability of microservice systems in practice. Throughout the implementation phase, careful attention is given to translating architectural designs into tangible systems. Validation activities are conducted to verify that the microservice architecture aligns with the intended objectives and requirements [SS02][SS16][SS27]. Once deployed, the system enters the operational phase, where it is actively used and monitored to ensure its smooth functioning [SS06]. Maintenance tasks, such as bug fixing, performance optimization, and feature updates, are performed to keep the microservice system in optimal condition [SS05][SS10][SS11][SS25][SS26]. Additionally, rigorous testing is carried out to verify the reliability, performance, and resilience of the architecture [SS32][SS34][SS35][SS36][SS37][SS41]. By emphasizing these essential implementation phases, organizations can effectively adopt microservice architectures and reap the benefits of scalability, flexibility, and maintainability offered by this architectural style.

• Standardizing microservice architecture, interfaces, and related aspects such as *load balancing*, *fault detection*, and *autoscaling* [SS08][SS34][SS43].

• Designing fault-tolerant and event-driven/asynchronous microservices, particularly for *smart systems*, *fog applications*, and *IoT applications* [SS05][SS10][SS38][SS42].

• Transitioning from specific solutions and their related validation to more general solutions through *fundamental research*, *reusable practices*, and *lessons learned* [SS20][SS25][SS31].

• Addressing the *complexities of the migration process from monolithic applications to microservice-based architectures* systematically [SS07][SS09][SS16][SS23], as well as tackling challenges related to *microservice identification*, *granulation*, and *proper design* [SS38].

• Conducting additional systematic literature reviews that consider gray literature to *compare findings and challenges identified in both white and gray literature* [SS14][SS15][SS16][SS25][SS31][SS33], extending existing secondary studies to *include the latest knowledge supplements* [SS07][SS15][SS29][SS40] or *exploring additional databases for comprehensive coverage* [SS20][SS40], and enhancing literature review approaches by *improving data extraction and synthesis methods, validity, and quality assessment of primary studies* [SS14][SS18][SS21].

All these research directions aim to advance our understanding of microservices architecture and facilitate the development of best practices, standardized approaches, and improved methodologies in this rapidly evolving field.

The analysis of the directions for future research reveals that they are numerous and diverse. This can be attributed to the relative novelty of microservices and microservice architectures, and their unique, yet relatively unexplored nature, characterized by heterogeneity, decentralization, and independence. Despite their established status and known design practices, challenges persist in effectively decomposing monolithic systems and adopting microservice architectures. Research studies suggest a myriad of migration strategies, technologies, tools, approaches, and languages for this purpose. Nevertheless, there is still room for improvement, clarification, and standardization in these areas. The breadth of future research directions signifies that the field of microservice analysis remains open to innovations and methodologies. These advancements hold the potential to support the development of more robust systems and architectural designs, particularly on an industry level. Since their introduction, the influence and impact of microservice architectures have significantly grown.

This study provides a valuable roadmap for researchers, highlighting areas that require further exploration. It also serves as a guide for practitioners, enabling them to assess the progress made thus far and determine which tools and approaches are suitable for practical implementation. Nonetheless, numerous challenges persist in this field, calling for continued breakthroughs in the years to come.

5. DISCUSSION

The following subsections present the implications of the study for researchers and industry practitioners, as well as validity considerations and recommendations for further research.

5.1. Implications

The research encompasses studies published within the last seven years. The analysis reveals a gradual increase in the number of papers focused on the topic of microservices, indicating its growing popularity and the increasing interest in this subject. Moreover, the realm of microservices covers a wide range of areas. The analyzed papers address various aspects, including the architecture, life cycle, migration, implementation, and challenges. This work's significance lies in its contribution to a comprehensive body of knowledge constructed through the examination of 44 secondary studies. Researchers and industry practitioners who aim to explore this field can utilize this knowledge as a foundation to delve into microservices further.

5.1.1. Implications for researchers. Researchers from academic institutions and PhD students can acquire valuable insights into cutting-edge research on microservices by carefully examining the paper's findings. The presented results, along with the comprehensive list of references, serve as an excellent starting point for researching specific microservices-related topics, such as microservice architecture, life cycle issues, migration, or technical implementation within specific domains. Furthermore, researchers can leverage the proposed methodology for conducting systematic literature reviews in their studies. It is worth noting that the research findings indicate a lax implementation of quality assessment measures in the evaluated papers. This observation highlights areas that require further investigation and emphasizes the need for more stringent quality assessment practices in future studies.

5.1.2. Implications for practitioners. The findings reveal the current research focus and the growing interest in various topics and areas. This information is significant for industry practitioners as it helps them stay informed about the most actively researched subjects. Practitioners can easily access information on specific topics related to microservice architecture, including models, patterns, granularity, quality attributes, life cycle issues, migration challenges, and technical implementation in specific domains. Furthermore, they can gain valuable insights into recent

methods, topics, tools, and technologies, which can be applied to enhance their professional work.

5.2. Validity

In empirical software engineering, it is crucial to mitigate threats to validity, which enhances the study's quality and ensures the reliability of the findings [31]. Validity assessment encompasses internal and external validity.

5.2.1. Internal validity. Our study acknowledges several threats to methodological validity, which should be treated as constraints since it is impossible to guarantee complete coverage. However, by adhering to our methods, we have minimized the potential for bias. The threats addressed in this study include the selection of search strings, digital library selection, selection criteria for secondary studies, and the construction of study findings. To address the first threat, we employed two search strings with relevant keywords aligned with our research objective, covering both SLRs and SMSs. For the second threat, we utilized six reputable digital libraries, encompassing publications from esteemed journals and international conference proceedings. We employed the snowballing technique to ensure a comprehensive collection of references. Regarding the third threat, we employed a five-phase approach to select primary studies, incorporating carefully defined inclusion and exclusion criteria, as well as quality assessment of secondary studies. The fourth threat was mitigated by conducting the research in a manner that allows for easy traceability of all findings back to the original secondary studies. Additionally, all authors actively participated in data extraction and analysis processes, minimizing the potential for misinterpretation of the initial data and the constructed findings. The detailed description of the research process in the third section of the paper provides transparency by allowing insights into all stages of the research, bolstering the validity and reliability of the constructed findings.

5.2.2. External validity. The generalizability of the findings is a key concern when assessing study external validity. As this study specifically focuses on the field of microservices as a tertiary study, the findings cannot be directly extrapolated to other domains of software engineering. Nonetheless, the detailed description of the research process and the construction of research findings enable the replication and adaptation of the procedure in similar studies within different areas of software engineering. Through such implementation and adaptation, the presented research process can be generalized and applied effectively.

5.3. Recommendations

This section provides comprehensive recommendations for further literature reviews in the field of microservices, encompassing primary, secondary, and tertiary studies. These recommendations stem directly from the research findings presented in this study. The primary objective is to address areas that have received limited attention in the analyzed secondary studies and offer guidance on enhancing existing research and enriching the knowledge base. The recommendations are organized and presented based on the overarching topics discussed in the research findings section, which include quality assessment trends, microservices architecture, life cycle issues, migration, technical implementation in different domains, and further research directions.

5.3.1. Quality assessment trends. According to guidelines for conducting literature reviews, quality assessment of included studies is typically considered a mandatory component of systematic literature reviews, whereas it may not be required for systematic mapping studies. Consequently, the quality assessment question Q4 received a low score of 0.37. However, it is important to note that the overall quality of the review study, particularly the quality of findings and study validity, is greatly influenced by the assessed quality of the studies included in the review. Therefore, the primary recommendation for addressing this issue is to assess the quality of all the studies selected for inclusion in the review. Additionally, it is advisable to establish a minimum value threshold for study inclusion to maintain rigorous standards throughout the review process.

5.3.2. Architecture. Research about microservices architecture exhibits a significant level of research maturity, particularly in the areas of architecture analysis, pattern utilization, and the adoption of various methods for architectural representation. Moreover, considerable attention has been given to exploring quality attributes and their associated trade-offs. However, the research findings indicate the necessity for additional primary and secondary studies to address certain aspects. Specifically, there is a need to evaluate quality attributes and metrics pertaining to granularity, runtime architecture visualization, as well as specific quality attributes such as context awareness, integrability, fault tolerance, upgrading, and availability. Conducting further studies in these areas will contribute to a more comprehensive understanding of microservices architecture and its associated quality considerations.

5.3.3. Life cycle issues. Researchers have dedicated substantial attention to life cycle issues, particularly the utilization of contemporary approaches like DevOps and continuous software engineering practices (continuous integration, continuous delivery, continuous deployment, and monitoring). However, certain topics have not received sufficient inquiry, including human factor issues, domain-driven design, and specific maintenance and operation concerns, such as load balancing and anomaly detection, which are closely associated with the system's availability and

reliability attributes. It is recommended that further research be conducted to address these areas and deepen our understanding of their impact on microservices.

5.3.4. Migration. The migration of legacy systems, typically characterized by monolithic architectures, to microservice architectures is a significant research topic. Numerous methods and tools have been proposed and evaluated in real-world settings, forming a substantial knowledge base for practitioners and researchers alike. However, upon analyzing the study findings, it is evident that two key areas, namely motivation and success factors, have not been sufficiently addressed in the existing research. Investigating these aspects requires a comprehensive examination of both technical and non-technical elements, encompassing organizational and human factors, in both empirical studies and literature reviews that systematically address these issues. By addressing these gaps, future research can provide valuable insights into the motivation and success factors associated with migrating legacy systems to microservice architectures.

5.3.5. Technical implementation in different domains. Limited research has been documented in research papers and literature reviews concerning the technical implementation of microservices across various domains. Most of the research focuses on the initial development of microservice-based systems, with limited evaluations conducted in real-world settings, primarily within IoT-based smart systems. Consequently, there is a lack of empirical evidence regarding the implementation of proposed solutions and the utilization of technologies in real systems. Furthermore, projects carried out by academic institutions often lack involvement from industry experts. There is a pressing need to conduct implementation studies within authentic settings across different domains to obtain valuable feedback on architecture models, methodologies, and employed technologies. It is also strongly recommended to engage industry experts to gain a more comprehensive understanding of microservice-based system implementations. By addressing these gaps, researchers can enhance the practical applicability and effectiveness of microservices in real-world scenarios.

5.3.6. Methodological issues for further research. Upon examining the methodological aspects of secondary studies, it is evident that there are several areas for further enhancement in review studies. The presented literature review offers recommendations in the following areas. (1) Integration of gray literature: it is recommended to incorporate gray literature into review studies to complement the insights obtained from white literature. This inclusion will provide a more comprehensive understanding of the industry experts' perspectives and the current state of practice. (2) Utilization of data extraction and synthesis methods from social sciences: to enhance the development of findings in a more structured manner, it is suggested to employ data extraction and synthesis methods commonly used in social sciences. These methods can include grounded theory [32] for developing theories, general inductive analysis [33] for creating catalogs with categories, or inductive thematic analysis [34] for constructing thematic frameworks, leading to the identification of patterns in practice [35], and the development of theories necessary for building and extending a knowledge base in this area of software engineering [36]. By implementing these recommendations, future review studies can further refine their methodologies and generate more robust and structured findings, contributing to the advancement of knowledge in the field.

6. CONCLUSIONS

This tertiary study offers a comprehensive analysis of recent research on microservices, focusing on publication trends, quality trends of published secondary studies, research topics, domains of implementation, and future research challenges. The findings stem from a systematic literature review process, which involved the selection of 44 secondary studies. The study addresses research implications for both academic researchers and software industry experts, discusses the validity of the study, and provides detailed recommendations for further research. The primary contribution of this study lies in its in-depth exploration of microservices implementation across various domains, as well as providing detailed and structured recommendations for further research endeavors.

Building upon the research findings, recommendations for further literature reviews include improving the quality assessment of analyzed studies, conducting more detailed reviews of architecture quality attributes, exploring human factor issues, and addressing maintenance and operation concerns. Additionally, conducting reviews that focus on the technical implementations of microservices within different business and human life domains is advised to gain a deeper understanding of their advantages, disadvantages, and comparative analysis with other architectural approaches. The inclusion of domain experts in the preparation and implementation of these literature reviews will lead to more precise and valid findings.

From the methodological standpoint, further reviews should consider incorporating qualitative social science methods, such as inductive thematic analysis or grounded theory, to obtain more structured and reliable findings. This approach may aid in identifying patterns in practice, and contribute to the development of theories necessary for building a knowledge base in this area of software engineering.

Future work will be directed toward three main areas. Firstly, there will be an emphasis on incorporating gray literature in the review process to gain insights

predominantly from industry experts and to compare the state of practice from both academic researchers' and industry experts' perspectives. Secondly, there will be a focus on conducting literature reviews specifically centered on microservices implementation in different business and human life domains. Lastly, a more comprehensive review of organizational issues and the human factor in the microservices life cycle will be undertaken, addressing areas that have received less attention in recent literature.

APPENDIX A

LIST OF SECONDARY STUDIES

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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